



Swinburne University of Technology Hawthorn Campus Department of Computing Technologies

COS20028 Big Data Architecture and Application Assignment 1 - Semester 2, 2024

Assessment Title: MapReduce Application.

Assessment Outcome: 20%

Due Date: AEST 23:59 on 30/08/2024.

Assessable Item:

- One (1) piece of a report containing the complete flowchart showing your design idea about how to prepare the data, the tools being used in all steps, and the report for answers to the given questions.
- One (1) piece of a video explaining how you found answers to the question, the challenges you encountered, and how you solved them.

Purpose of Assignment

This assignment tests whether a student is capable of using MapReduce to cope with real-world problems and achieve a specific goal. The solution designed should be reasonable, practical, and manageable.

Introduction

MapReduce allows for relatively fast and easy processing over very large datasets using a cluster of commodity machines. In this assignment, students will become more familiar with and gain practical experience with the MapReduce Programming Model on top of the Hadoop software platform. This assignment requires students to understand the process of designing, setting up, and executing MapReduce tasks over the given dataset on a single node. For this assignment, students are required to run Hadoop on the virtual machine and solve the tasks given. The student will be given the task of implementing their own MapReduce job and analysing the outcome produced. Students are required to place comments along with the code for readability. Explaining your design and how you get the answers to each question in the video are essential.

Dataset

The dataset source has been prebuilt into your Virtual Machine Image. You can find the source file at the following path: /home/training/workspace/2024_assignment1. The source file name is "austlang_dataset.csv". If you fail to locate the dataset, you can use the browser in the virtual machine and download it from Canvas.

The given dataset is a collection of Australian First Peoples' languages. It has been organised in the format of a table with multiple columns.

Assignment Goal

This assignment aims to train students to analyse the problems they encounter and find the most suitable way(s) for accomplishing the given tasks in the real-world big data processing environment. Students will encounter some research and discover components to learn skills and knowledge from the project.

Assignment Task

Create a Java Project named `Assignment1` to produce a working Hadoop project, which will be used to answer the questions below. Prepare a short video (no more than 3 minutes) to explain how you designed your solution, the challenges you encountered, how you found the solutions for them, and how you found the answers to each question.

Your codes must fulfil the following criteria and can be used to find answers to the questions:

1. Correct implementation of the Mapper class(es) with readability and well-structured codes and comments. **[20 marks]**
2. Correct implementation of the Reducer class(es) with readability and well-structured codes and comments. **[20 marks]**
3. Correct implementation of the Driver class(es) with readability and well-structured codes and comments. **[20 marks]**
4. List all commands used in this assignment in order, as well as readable execution screenshots with your name and student ID printed from the driver code. **[15 marks]**
5. You are assigned to find the word counts of any terms in the given dataset starting with the upper-case letter "A," "C," "G," "W," or "Y," followed by lowercase letters regardless of the length. Answering the questions:
 - a) What is the term under the given criteria with the highest appearance count, and what is its count?
 - b) What is the term under the given criteria for the second high appearance count, and what is its count?
 - c) How many terms only have a single count in the document?**[25 marks]**
6. Video presentation with the required content. **[Pass/Fail]**

Please use the template to prepare your submission.

Hints and Useful Information

1. It is essential to use regular expressions in your design to capture the required terms in your source file.

2. Importing “java.util.regex.Matcher” and “java.util.regex.Pattern” will enable you using the regular expression for matching in the program. Use “Pattern.compile()” to set your pattern, and then use the “.match()” function and the “.find()” function to determine whether the desired pattern appears.
3. You may need to use more than one MapReduce code to complete the task.
4. If you use “less” as the document reader, the search function can be triggered by pressing “/” on the keyboard to assist you in finding the desired item.
5. **You are required to reveal yourself in the video as evidence that you made it yourself.**

Submission Requirement

Only the submission using the given report template is acceptable. The video for the presentation should be within three minutes.

Failure to adhere to the submission requirements will immediately result in a failure for this assignment.

Rubric: Assignment 1

	Task	Mark
1.	Task 1	20 marks
2.	Task 2	20 marks
3.	Task 3	20 marks
4.	Task 4	15 marks
5.	Task 5	25 marks
6.	Task 6	Pass/Fail

----- End of Assignment -----